TELECOM CUSTOMER CHURN ANALYSIS

Data Analysis Project Report

Excel • Power Query • PivotTables • Power BI • DAX

Project focus	Customer base, usage, revenue and churn analysis
Primary source	Telecommunications subscriber dataset
Analytical tools	Microsoft Excel, Power Query and Microsoft Power BI
Reporting structure	Four-page interactive Power BI report

This report records the analytical work completed from raw data through dashboard delivery. It is written as a project record rather than as a tutorial or set of instructions.

1. Project Background

The project examined a telecommunications subscriber dataset to develop a clear, interactive view of customer composition, service usage, revenue and churn. The work was completed as an end-to-end data analysis workflow, beginning with an audit of the raw workbook and ending with a four-page Power BI dashboard.

The analysis combined spreadsheet-based exploratory analysis with Power Query transformations, a relational data model, DAX measures and interactive visual reporting. The final report separated descriptive findings from assumptions about causality so that observed patterns were not presented as proven causes of churn.

2. Business Problem

The available subscriber records contained customer, plan, regional, usage, billing and churn information, but the raw data required preparation before it could be used reliably for analysis. The project therefore addressed two related needs: establishing a clean analytical dataset and using that dataset to describe the customer base and identify observable churn patterns.

The completed analysis addressed the following business questions:

- What is the size and composition of the active and churned subscriber base?
- How are subscribers distributed across regions, plans and account types?
- How do customers use data, voice and SMS services?
- How is monthly revenue distributed across plans, regions and account types?
- How does observed churn vary by tenure, plan, region, account type, payment method and complaints?

3. Dataset and Source Structure

The working Excel workbook contained four principal sheets: Project_Brief, Plans, Regions and Subscribers_RAW. The Plans and Regions sheets functioned as lookup tables, while Subscribers_RAW contained the subscriber-level records used for the main analysis.

Field	Role in analysis
Subscriber ID	Unique customer identifier used for subscriber counts and duplicate checks
Region ID / Plan ID	Keys linking subscriber records to regional and plan lookup tables
Account Type	Customer account classification
Tenure (Months)	Length of the subscriber relationship
Data Usage (GB)	Customer data consumption
Voice Minutes	Customer voice usage
SMS Count	Customer SMS usage
Complaints (90 Days)	Recorded complaints during the 90-day period
Monthly Bill (GHS)	Monthly customer billing amount used for revenue analysis
Churned	Subscriber churn status
Payment Method	Payment method recorded for the subscriber

4. Data Audit and Cleaning

The first stage was an audit of the raw subscriber table. The audit identified 706 raw rows and 700 unique Subscriber IDs, indicating six duplicate subscriber records. The audit also revealed inconsistent text formatting, including values such as " prepaid " and "Prepaid". These inconsistencies were important because categorical analysis depends on consistent category values.

The raw data was then prepared through Power Query rather than by altering the original source table manually. Text fields were cleaned and standardized, data types were validated, duplicate subscriber records were addressed, and lookup identifiers were checked against the Plans and Regions tables. The raw source was retained separately from the transformed analytical data.

Power Query provided a repeatable transformation layer between the source workbook and the reporting model. Selecting Close & Apply loaded the transformed data into Power BI without overwriting the original Excel worksheet.

The resulting preparation stage established a cleaner dataset for subsequent PivotTable analysis and Power BI modelling.

5. Exploratory Analysis in Excel

Before building the final dashboard, the cleaned subscriber data was examined through PivotTables. The analysis covered revenue, subscriber status, complaints and churn, tenure and churn, usage by plan, revenue by plan and regional churn.

5.1 Subscriber Status

The analysis recorded 667 active subscribers and 33 churned subscribers. Based on 700 unique subscribers, the observed overall churn rate was approximately 4.71%. This provided the baseline against which segment-level churn rates were interpreted.

5.2 Churn by Plan

Plan	Observed churn rate
Basic	3.06%
Business	4.24%
Premium	6.06%
Flex	9.09%

The PivotTable showed variation in observed churn rates across the four plans. These differences describe the dataset and do not by themselves establish that plan choice caused churn.

5.3 Complaints and Churn

Complaints in 90 days	Churned subscribers	Observed churn rate
0	14	3.5%
1	8	6.3%
2	6	10.3%
3	2	14.3%
4+	3	16.7%

The observed churn rate increased across the complaint groups in this dataset. The pattern was therefore treated as an association requiring further investigation, rather than as evidence that complaints directly caused customers to leave.

5.4 Revenue by Plan

Plan	Total monthly revenue (GHS)
Basic	19,874.25
Business	46,417.17
Premium	17,872.88
Flex	22,529.63

The Business plan generated the largest total monthly revenue within the four-plan dataset. Revenue was considered alongside subscriber volume and churn rather than as a standalone measure of customer performance.

6. Data Model

The cleaned subscriber table was structured as the central fact table, with Plans and Regions serving as dimensions. The model connected plan and regional identifiers through one-to-many relationships.

Relationship	Cardinality	Filter direction
Plans[Plan ID] → Subscribers_RAW[Plan ID]	1 : *	Single
Regions[Region ID] → Subscribers_RAW[Region ID]	1 : *	Single

This model separated descriptive lookup information from subscriber-level records and provided a consistent foundation for Power BI calculations. Relationships were kept at the identifier level to avoid unnecessary duplicate paths through both IDs and descriptive names.

7. Power BI Measures and Analytical Layer

After the model was established, reusable DAX measures were created for the main KPIs and analytical dimensions used in the report. Measures were preferred for recurring calculations so that the same definitions could respond consistently to filters and slicers across report pages.

7.1 Core Measures

Total Subscribers

```
Total Subscribers =
DISTINCTCOUNT(Subscribers_RAW[Subscriber ID])
```

Churned Subscribers

```
Churned Subscribers =
CALCULATE(
[Total Subscribers],
Subscribers_RAW[Churned] = "Yes"
)
```

Active Subscribers

```
Active Subscribers =
CALCULATE(
[Total Subscribers],
Subscribers_RAW[Churned] = "No"
)
```

Churn Rate

```
Churn Rate =
DIVIDE(
[Churned Subscribers],
[Total Subscribers],
0
)
```

Total Monthly Revenue

```
Total Monthly Revenue =
SUM(Subscribers_RAW[Monthly Bill (GHS)])
```

ARPU

```
ARPU =
DIVIDE(
[Total Monthly Revenue],
[Total Subscribers],
0
)
```

7.2 Supporting Measures

Additional measures were created to support the customer-behaviour and churn views, including Average Data Usage, Average Voice Minutes, Average SMS Count, Average Monthly Bill, Average Complaints and Churned Revenue. These measures allowed the same underlying fields to be presented consistently across different report contexts.

8. Dashboard Development

The final Power BI report was organized into four pages rather than placing all visuals on a single canvas. Each page was given a distinct analytical purpose so that users could move from the overall customer base to customer behaviour, revenue and finally churn.

8.1 Overview

The Overview page presented the overall state of the subscriber base. KPI cards displayed Total Subscribers, Active Subscribers, Churned Subscribers, Churn Rate and Total Monthly Revenue. Supporting visuals described subscribers by region, plan and account type, with customer status shown as a composition view. Slicers were included to support interactive filtering.

8.2 Customer Behavior

The Customer Behavior page focused on service usage. It included average data usage, average voice minutes and SMS activity by plan, together with a Data Usage versus Monthly Bill relationship view. Payment method and complaint distributions were also included to provide additional customer-context information.

8.3 Revenue

The Revenue page focused on the financial structure of the subscriber base. KPI cards presented Total Monthly Revenue, ARPU, Total Subscribers and Average Monthly Bill. Revenue was analyzed by plan, region and account type, while ARPU by plan and the distribution of monthly bill levels provided additional financial context.

For the bill distribution analysis, monthly bills were grouped into ranges such as 0–50, 51–100, 101–150, 151–200, 201–250 and 251+. A numeric sort field was used so that the ranges appeared in logical order rather than alphabetical order.

8.4 Churn

The Churn page concentrated on customer retention patterns. It displayed Total Subscribers, Churned Subscribers, Churn Rate and Average Complaints, followed by churn-rate comparisons across tenure, region and plan. Complaints were examined alongside churn counts and rates, and additional segmentation was provided for account type and payment method.

9. Report Navigation and Interaction

The four-page report was connected through a persistent sidebar. A Power BI Page Navigator was used to provide navigation between Overview, Customer Behavior, Revenue and Churn. The navigation design was carried across the report pages so that the report retained a consistent structure while the analytical content changed.

Interactive elements such as slicers and visual cross-filtering were used to allow the dashboard user to examine the same measures across different customer segments without creating separate static reports.

10. Analytical Findings

The completed analysis produced several descriptive findings from the available snapshot of subscriber data.

- Overall churn: 33 of 700 unique subscribers were recorded as churned, corresponding to an observed churn rate of approximately 4.71%.
- Plan variation: observed churn rates differed across Basic, Business, Premium and Flex plans.
- Complaint pattern: the complaint groups showed progressively higher observed churn rates, from 3.5% among customers with no recorded complaints to 16.7% among customers with four or more complaints.
- Revenue contribution: the Business plan generated GHS 46,417.17 in total monthly revenue in the analyzed dataset, compared with GHS 22,529.63 for Flex, GHS 19,874.25 for Basic and GHS 17,872.88 for Premium.

- Segmentation matters: churn counts and churn rates provide different information. Segment size was therefore considered when interpreting subgroup results.

These findings describe patterns in the observed dataset. They were not interpreted as proof that a particular plan, tenure level or complaint level caused churn.

11. Business Implications

The analysis provides a structured basis for further retention investigation. The variation in churn across plans, tenure groups, regions and complaint levels indicates that churn should be monitored at segment level rather than through a single overall KPI. The revenue analysis also shows why customer value and churn need to be considered together when evaluating retention priorities.

The dashboard can therefore serve as a monitoring and exploratory tool: management can filter the customer base, inspect segment-level churn and revenue, and identify areas where additional customer-level or longitudinal analysis would be useful. The report does not assume that an observed relationship is sufficient evidence for a retention intervention.

12. Limitations

- The dataset represents a snapshot rather than a complete longitudinal customer history.
- Only 33 churned subscribers were observed, which limits the stability of detailed subgroup comparisons.
- Six duplicate subscriber records were identified in the raw data and required explicit handling during preparation.
- The available variables do not establish the causal drivers of churn.
- The dataset does not provide enough historical information to quantify the long-term financial impact or lifetime value associated with churn.

13. Conclusion

The project progressed from raw subscriber records to a structured analytical dataset, exploratory PivotTables, a relational Power BI model, reusable DAX measures and a four-page interactive dashboard. The completed report provides a consolidated view of customer composition, service usage, revenue and churn within the available telecommunications dataset.

The most important outcome of the project is the separation of data preparation, analysis and communication into a reproducible workflow. The resulting dashboard makes the observed patterns accessible for further investigation while maintaining a clear distinction between descriptive evidence and causal conclusions.

14. Project Workflow Record

Stage	Completed work
Business understanding	Defined the customer, usage, revenue and churn questions addressed by the analysis.
Raw-data audit	Examined row counts, unique subscribers, duplicates and category consistency.
Data preparation	Applied Power Query cleaning, standardization, type validation and duplicate handling.
Validation	Checked lookup identifiers against Plans and Regions.

Stage	Completed work
Exploratory analysis	Built Excel PivotTables covering revenue, status, usage, complaints, tenure and churn.
Data modelling	Established Plans and Regions as dimensions linked to the subscriber fact table.
DAX layer	Created reusable subscriber, churn, revenue and usage measures.
Dashboard development	Built four analytical Power BI pages with KPI cards, charts, slicers and navigation.
Interpretation	Documented observed patterns and separated descriptive evidence from causal claims.
Presentation	Structured the report and dashboard for interactive exploration and future analysis.

Appendix A — Key DAX Definitions

The following measures formed the core calculation layer used in the report.

Average Data Usage

```
Average Data Usage =
AVERAGE(Subscribers_RAW[Data Usage (GB)])
```

Average Complaints

```
Average Complaints =
AVERAGE(Subscribers_RAW[Complaints (90 Days)])
```

Average Voice Minutes

```
Average Voice Minutes =
AVERAGE(Subscribers_RAW[Voice Minutes])
```

Average SMS Count

```
Average SMS Count =
AVERAGE(Subscribers_RAW[SMS Count])
```

Average Monthly Bill

```
Average Monthly Bill =
AVERAGE(Subscribers_RAW[Monthly Bill (GHS)])
```

Churned Revenue

```
Churned Revenue =
CALCULATE(
[Total Monthly Revenue],
Subscribers_RAW[Churned] = "Yes"
)
```

Document status: completed project documentation based on the established telecom churn analysis workflow and results.